ReFood Connect — Quality Assurance & Testing Evidence Portfolio

Digital Assignment 3 (Part A Evaluation) | Software Engineering (BCSE301L)

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Official GitHub Repository: https://github.com/ChiragAsija1323/ReFood-Connect
Verification Milestone: Review 3 / DA-3 Single Compiled Quality PDF

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1. Quality Verification Executive Summary

Verification Target	Observed Result	Compliance Status
Integration & Unit Suites	27/27 Tests Passed (100% Pass Rate)	VERIFIED (PASS)
Test Execution Time	5.32 seconds (In-Memory SQLite Isolation)	OPTIMAL
Mutation Testing Tool	StrykerJS v8.7 with Jest Runner	VERIFIED
Baseline Mutation Score	60.31% (79 Killed, 15 Survived, 37 Uncovered)	RECORDED
Hardened Mutation Score	**84.73%** (111 Killed, 14 Survived, 6 Uncovered)	EXCELLENT (+24.4%)
Frontend Production Build	Vite 8.1.5 build completed in 1.61 seconds	VERIFIED (PASS)
Docker Compose Specification	Multi-container config verified (client, server, db)	VERIFIED (PASS)
RBAC & State Machine Integrity	Admin dashboard, volunteer transit guards enforced	VERIFIED (PASS)

2. Automated Integration & Regression Testing Evidence

The test harness combines Jest, Supertest, and SQLite memory mode to execute tests with zero environmental dependencies.

```
$ cd server && npm test
> cross-env NODE_ENV=test jest --detectOpenHandles --forceExit
PASS __tests__/auth.test.js (5 tests passed)
PASS __tests__/donations.test.js (7 tests passed)
PASS __tests__/regression.test.js (7 tests passed)
PASS __tests__/donationService.test.js (8 tests passed)
Test Suites: 4 passed, 4 total | Tests: 27 passed, 27 total | Time: 5.323 s
```

3. StrykerJS Mutation Testing Evidence & Analysis

StrykerJS was configured to mutate server/services/donationService.js to test state machine transitions and boundary validations.

Metric	Baseline Run	Hardened Run	Analysis / Outcome
Total Mutants Evaluated	131 mutants	131 mutants	Full AST syntax coverage
Mutants Killed (✓)	79 mutants	**111 mutants**	+32 mutants caught (+40.5%)
Mutants Survived (✘)	15 mutants	14 mutants	Non-critical edge logic
Uncovered Mutants (∅)	37 mutants	**6 mutants**	Dropped by 83.8%
Covered Mutation Score	84.04%	**88.80%**	High mutation accuracy
Total Mutation Score	60.31%	**84.73%**	**Grade A Verification (+24.42%)**

4. System Building & Version Management Evidence

Frontend Build (Vite 8.1.5):

```
vite build
dist/index.html (0.46 kB) | dist/assets/index.js (296.81 kB gzip: 95 kB)
✓ built in 1.61s
```

Docker Compose Orchestration:

```
docker compose config
Services: client (5173), server (8000), db (mysql:8.0:3306 with healthcheck ping)
Status: Validation successful (Exit code 0)
```

Version Management (Git Log History):

```
* a038c11 [DA-3] Implement Admin governance, volunteer status transitions, test harness, Stryker
mutation testing
* 55da35f Update UI screenshots with higher resolution images
* 3a12bb3 Final README cleanup
* c14c4f8 Update README with accurate DA-2 software design details and tech stack
* 96d28d4 Embed Architecture Diagram and UI screenshots directly into Design Document
* 1074b67 DA-2: Add complete Architecture Draw.io, UI wireframes, and Software Design Document
```

5. Functionalities & Screenshot Gallery Guide

The visual proof checklist confirms working execution across all 5 user roles:

Figure ID	Component / Screen	Demonstrated Behavior
Figure 1	Terminal / Jest Execution	27/27 integration and regression tests passing
Figure 2	Stryker Mutation Report	84.73% mutation score on donationService.js
Figure 3	Git Version Management	Linear chronological commit history from DA-1 to DA-3
Figure 4	Vite Production Build	Production client build completing in 1.61 seconds
Figure 5	Docker Compose Validation	Orchestration syntax and healthcheck validation
Figure 6	User Registration	Role-based account registration with password hashing
Figure 7	Food Donor Dashboard	Creating surplus listing in 'Available' status

Figure 8	NGO Dashboard	Discovering and requesting meals (Status -> Requested)
Figure 9	Volunteer Dashboard	Accepting task and updating transit (-> Delivered)
Figure 10	Administrator Dashboard	Live platform statistics, audit table, registered users